a Ingestion and biophysical gate

VIIRS 375 m
Suomi-NPP · NOAA-20/21

ERA5-Land
3-hourly → daily · 0.25°

MapBiomas C2
annual · prior-year join

Hansen GFC
tree-cover loss

Historical priors
causal · 3-year window

daily 1-km² stack →

Biophysical gate

Fuel = 1

VPD ≥ 0.8 kPa

$P_{48h} < 10$ mm

excludes non-combustible and wet cell-days

↓

Gated candidate cell-days

21.7% cell-days removed

b D2 target operator

$F_t = \{i:\ \text{fire}(i,t) = 1\}$

$I_{i,t} = 1 \Leftrightarrow i \in F_t,$ $I \cap P = \emptyset$

$N_{25}(i, t-1) = \emptyset,\ H_{10}(i,t) = 0$

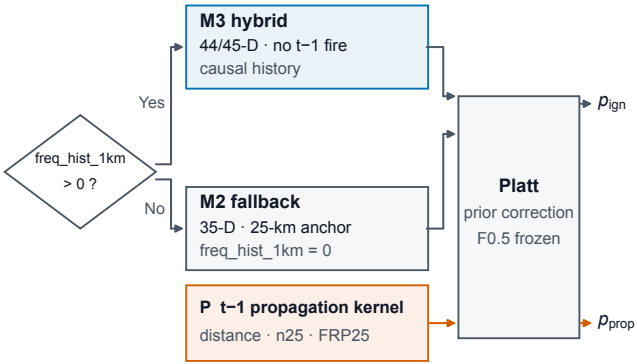
$P_{i,t} = 1 \Leftrightarrow i \in F_t,$ 25 km · 10 d

$N_{25}(i, t-1) \neq \emptyset$

I primary ignition | **P tactical propagation**

domains disjoint

c Decoupled inference

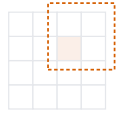


d Auditable evaluation

Temporal split

2018–2022	2023	2024–2025
Train	Val	Test

Spatial holdout



50-km blocks
20-km buffer
spatial only

Uncertainty

date-grouped
bootstrap
500 CI · 1,000 paired

Decision value

cost-loss

Open code + data

reproducible release